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标题	Hydrogenation of Alkynes and Olefins Catalyzed by Quaternary Ammonium Salts
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摘要（原文节选）

AbstractCatalytic hydrogenation of unsaturated hydrocarbons to alkenes and alkanes using molecular hydrogen is one of the most fundamental transformations in organic synthesis. While methodologies involving transition metals as catalysts in homogeneous and heterogeneous processes have been well developed, metal‐free catalytic hydrogenation offers an ideal approach for future chemistry. Herein, the common and inexpensive quaternary ammonium salts are first introduced as catalysts in the catalytic hydrogenation syste ⋯

数据表 · Table 1（16 行，含数值行 16 行）

Optimization of hydrogenation conditions of alkynes to olefins.

Entry a)	R4NX	Pressure (MPa)	Solvent	Yield [%]
1	–	5.0	Toluene	ND
2	Bu4NCl	5.0	Toluene	93
3	Bu4NCl	0.5	Toluene	91
4	Bu4NCl	balloon	Toluene	48
5 b)	Bu4NCl	5.0	Toluene	39
6	Bu4NCl	0.5	THF	92
7	Bu4NCl	0.5	DCE	89
8	Bu4NCl	0.5	1,4‐Dioxane	86
9	Bu4NCl	0.5	DMF	78
10 c)	Bu4NCl	0.5	THF	87
11	Bu4NBr	0.5	THF	88
12	Bu4NI	0.5	THF	68
13	Bu4NBF4	0.5	THF	61
14	Bu4NOTf	0.5	THF	Trace
15	Me4NI	0.5	THF	Trace
16	Me4NCl	0.5	THF	Trace

脚注（逐字）：a) All reactions were performed with 1a (0.2 mmol) in 2 mL solvent at 80 ° C for 72 h. Yields are those for the isolated products; b) Reaction ran 48 h under 5 MPa; c) Reaction was carried out at 60 ° C.

正文对应句（逐字）：We started off our investigations using the terminal alkyne (1a) as the starting substrate to explore the optimal reaction conditions. The preliminary experiment was carried out in toluene under the pressure of H2 (0.5 MPa) at 80 ° C in the absence of quaternary ammonium salts (entry 1). As anticipat

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